

AI MACHINE LEARNING

# The Hunger Games: Hiring Edition

## Which Features Are Most Predictive of Hiring Outcomes?

Education · Experience · Skills · The strongest signals survive

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## THE SIGNALS

# Which Signals Actually Predict Employment?

Employment outcomes are shaped by education, experience, and technical skills — but not all signals are equally powerful. This project investigates which feature groups truly drive employability prediction and whether semantic skill embeddings outperform traditional background information.

## Why It Matters

Understanding which signals drive employability can improve hiring analytics, workforce research, and career guidance.

## Why It's Hard

Developer survey data is noisy, high-dimensional, and contains complex relationships between skills, education, and experience.

## The Goal

Determine which feature representations provide the strongest predictive power for employment outcomes.



#### PREPARATION

## Preparing the Tributes

### District Education

EdLevel (ordinal)

Gender/Age/MainBranch (one-hot)

Accessibility/MentalHealth (one-hot)

*handle\_unknown='ignore'*

### District Experience

YearsCode + YearsCodePro (StandardScaler)

Numeric values centered and scaled

Prevents regularization bias

### District Skills — 4 Weapons

Binary: sparse indicator

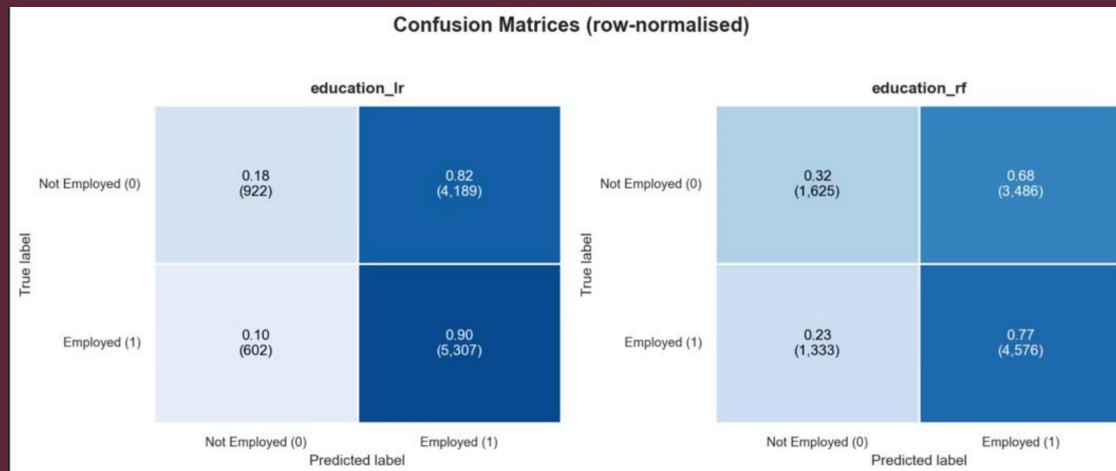
Counts: distinct skill count

Grouped: 8 tech-domain categories

Embeddings: all-MiniLM-L6-v2, 384 dimensions

① Embeddings capture semantic similarity

# Education Enters the Arena



## Logistic Regression

**F1:** 0.689

**ROC-AUC:** 0.564

## Random Forest

**F1:** 0.655

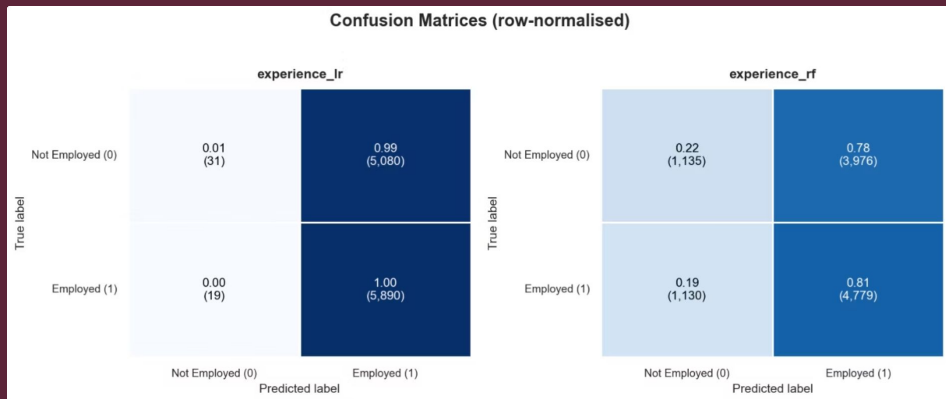
**ROC-AUC:** 0.567



### Key Takeaway

Education alone is a weak predictor of employability.

# Experience — A Critical Failure



1

## What Happened?

The model predicted almost everyone as employed, showing extremely weak class separation.

2

## Why Did It Fail?

YearsCode and YearsCodePro alone did not contain enough signal to reliably distinguish employment outcomes.

3

## Key Insight

The model's **ROC-AUC fell near 0.5**, indicating near-random class separation. Raw experience was far less predictive than semantic skills information and embeddings.

0.49

LR  
ROC-AUC

0.52

RF  
ROC-AUC



Degenerate prediction behavior detected — eliminated from the arena.

## Skills Enter the Arena

### 4 Skill Representations

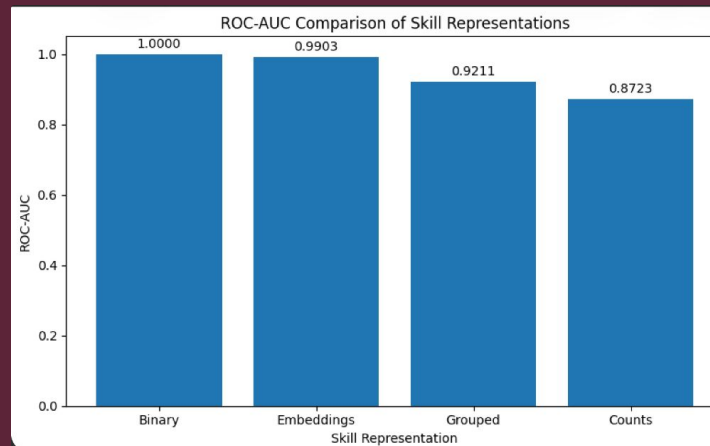
Counts: number of skills only  
Grouped: broad tech areas  
Embeddings: semantic skill profile  
Binary: exact skill fingerprint

| Model         | Accuracy      | Precision     | Recall        | F1            | ROC-AUC       |
|---------------|---------------|---------------|---------------|---------------|---------------|
| counts_lr     | 0.7803        | 0.8174        | 0.7600        | 0.7877        | 0.8723        |
| counts_rf     | 0.7863        | 0.7874        | 0.8238        | 0.8052        | 0.8723        |
| grouped_lr    | 0.8341        | 0.8415        | 0.8509        | 0.8462        | 0.9211        |
| grouped_rf    | 0.8140        | 0.8228        | 0.8323        | 0.8275        | 0.8966        |
| embeddings_lr | <b>0.9472</b> | <b>0.9470</b> | <b>0.9550</b> | <b>0.9510</b> | <b>0.9903</b> |
| embeddings_rf | <b>0.8867</b> | <b>0.8953</b> | <b>0.8930</b> | <b>0.8942</b> | <b>0.9607</b> |
| binary_lr     | <b>1.0000</b> | <b>1.0000</b> | <b>1.0000</b> | <b>1.0000</b> | <b>1.0000</b> |
| binary_rf     | <b>0.9946</b> | <b>0.9934</b> | <b>0.9966</b> | <b>0.9950</b> | <b>0.9999</b> |

Skills dominated the arena: counts helped, grouped skills improved, embeddings captured semantic depth, and binary skills achieved perfect separation.

# Binary Skills: Perfect Scores, Practical Problem

| Metric    | Binary LR |
|-----------|-----------|
| Accuracy  | 1.000     |
| Precision | 1.000     |
| Recall    | 1.000     |
| F1        | 1.000     |
| ROC-AUC   | 1.000     |



## Perfect Metrics Raised Concerns

Exact Skill Fingerprints  
Binary created one yes/no column per skill, making applicants highly separable.

## Shuffled-Label Test

Shuffled-label test dropped to random(0.489) → no obvious pipeline leakage

## Excluded from Combined Model

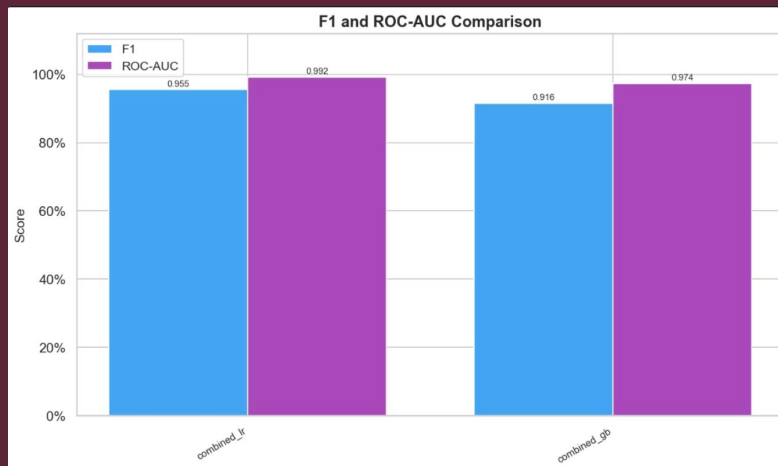
Excluded from combined model to avoid dominance

## Final Decision

Decision: Use embeddings for the final combined model.

Retain strong skill-based predictive power while being more generalizable and suitable for the combined model.

# Combined Models — The Final Showdown



## The Verdict

The combined embedding-based model achieved the strongest performance. The combined embedding-based Logistic Regression model achieved the strongest overall performance, outperforming the more complex Gradient Boosting model.

### Why Did Logistic Regression Win?

Semantic skill embeddings already separated the classes extremely well, making the problem close to linearly separable.

### Key Insights

High-quality embeddings reduced the need for complex nonlinear models.

Skills carried most of the predictive signal, while education and experience added only limited improvement.

Semantic skill embeddings proved far more powerful than traditional background features.

# Lessons From the Arena



## What Worked

- Skills, especially semantic embeddings
- Combined Logistic Regression
- Representations capturing semantic meaning



## What Failed

- Experience-only models
- Education-only models



## What We Learned

- Model performance isn't just about the highest score
- Perfect metrics can signal overfitting
- Feature representation often outweighs model complexity



# Skills Lead — Combined Model Wins

## SKILLS DOMINATE

Exact and semantic skill profiles carried the strongest signal

## EMBEDDINGS ADD VALUE

Strong performance without exact skill fingerprinting

## CREDENTIALS FALL SHORT ALONE

Education and experience were weak as standalone predictors

## FUTURE WORK

- Temporal validation
- Better experience features
- External hiring data